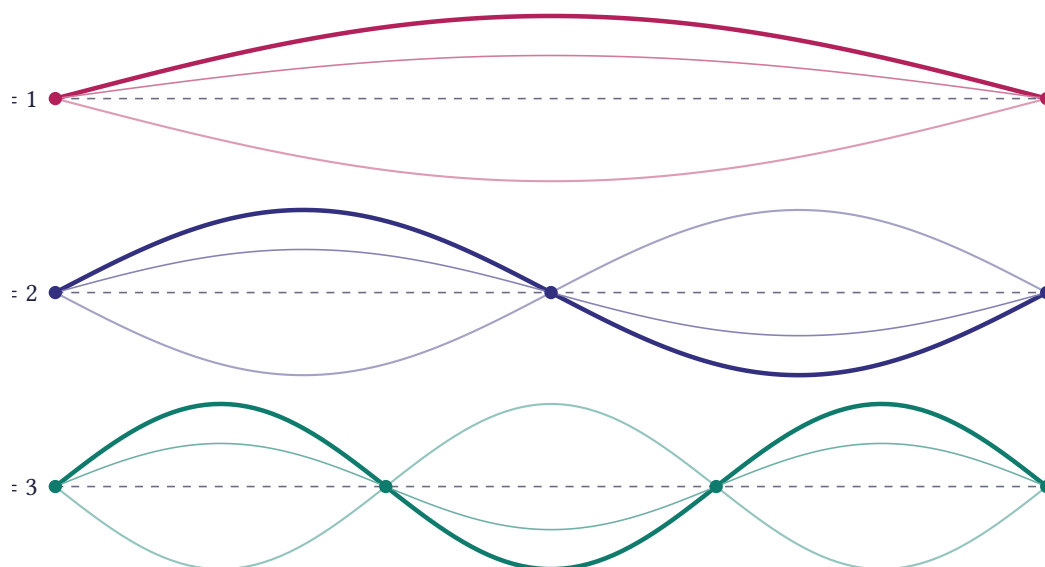


The One-Dimensional Wave Equation and Separation of Variables

معادله موجی یک بعدی و روش جداسازی متغیرها



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11.1 Introduction

Definition 11.1 ■ Differential equation

An equation in which a function and its derivatives appear.

Definition 11.2 ■ Ordinary differential equation

An equation written in terms of only one independent variable, with the derivatives of the function taken only with respect to that variable.

Definition 11.3 ■ Partial differential equation

A differential equation in which the function depends on several independent variables and the partial derivatives of the function with respect to the different variables appear in the equation.

The wave equation:

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad \frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \quad \frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

The heat equation:

$$\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad \frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right), \quad \frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$$

Laplace's equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \quad \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = 0, \quad \nabla^2 u = 0$$

11.2 The One-Dimensional Wave Equation

Modelling a vibrating string (deriving the equation governing $u(x, t)$).

Physical assumptions:

- The mass is distributed uniformly along the string (a homogeneous string), and the string is perfectly flexible.
- The string is stretched so tightly that the force of gravity on it is negligible.
- The deflection of the string from equilibrium is very small.

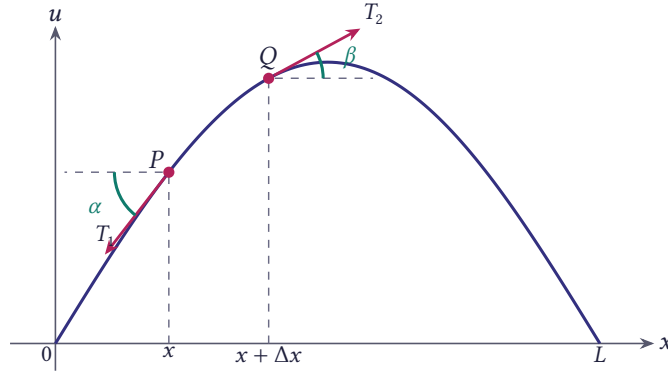


Figure 11.1 A segment PQ of the vibrating string: the tensions T_1 and T_2 at its ends, at angles α and β

$$T_1 \cos \alpha = T_2 \cos \beta = T = \text{const.}$$

$$T_2 \sin \beta - T_1 \sin \alpha = \rho \Delta s \frac{\partial^2 u}{\partial t^2}$$

where ρ is the mass density of the string and Δs the differential of the string length. Dividing by T :

$$T(\tan \beta - \tan \alpha) = \rho \Delta s \frac{\partial^2 u}{\partial t^2} \implies T \left(\frac{\partial u}{\partial x} \Big|_{(x+\Delta x, t)} - \frac{\partial u}{\partial x} \Big|_{(x, t)} \right) = \rho \Delta s \frac{\partial^2 u}{\partial t^2}$$

By the mean value theorem ($f(b) - f(a) = f'(l)(b - a)$, $l \in [a, b]$):

$$T \left(\frac{\partial u}{\partial x} \Big|_{(x+\Delta x, t)} - \frac{\partial u}{\partial x} \Big|_{(x, t)} \right) = T \frac{\partial^2 u}{\partial x^2} \Big|_{(l, t)} \Delta x = \rho \Delta s \frac{\partial^2 u}{\partial t^2}, \quad x < l < x + \Delta x$$

By assumption $\lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta s} = 1$, so

$$T \frac{\partial^2 u}{\partial x^2} = \rho \frac{\partial^2 u}{\partial t^2}, \quad c^2 \triangleq \frac{T}{\rho} \implies \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

For a unique solution of this equation we need boundary conditions and initial conditions:

- Boundary conditions: $u(0, t) = u(L, t) = 0$
- Initial conditions: $u(x, 0) = f(x)$ and $u_t(x, 0) = g(x)$, where $u_t \triangleq \frac{\partial u}{\partial t}$

11.3 Solving the One-Dimensional Wave Equation

$$\begin{cases} \frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2} \\ u(0, t) = u(L, t) = 0 \\ u(x, 0) = f(x) \\ u_t(x, 0) = g(x) \end{cases}$$

Here we solve the equation by **separation of variables**. Assume

$$u(x, t) = X(x) \times T(t)$$

$$\frac{\partial^2 u}{\partial x^2} = X''(x)T(t), \quad \frac{\partial^2 u}{\partial t^2} = X(x)T''(t) \implies X(x)T''(t) = c^2 X''(x)T(t) \implies \frac{T''}{c^2 T} = \frac{X''}{X} = k$$

k must necessarily be a negative number. Examining the cases:

If $k = 0$:

$$\frac{X''}{X} = 0 \Rightarrow X'' = 0 \Rightarrow X(x) = Ax + B, \quad X(0) = 0 \Rightarrow B = 0, \quad X(L) = 0 \Rightarrow A = 0 \Rightarrow X(x) = 0$$

If $k > 0$ ($k = \lambda^2$):

$$\frac{X''}{X} = \lambda^2 \Rightarrow X'' - \lambda^2 X = 0 \Rightarrow X(x) = Ae^{\lambda x} + Be^{-\lambda x}, \quad X(0) = 0 \Rightarrow A + B = 0, \quad X(L) = 0 \Rightarrow Ae^{\lambda L} + Be^{-\lambda L} = 0$$

$$\Rightarrow A = B = 0 \Rightarrow X(x) = 0$$

Hence k must be negative ($k = -\lambda^2$):

$$\frac{T''}{c^2 T} = \frac{X''}{X} = -\lambda^2 \Rightarrow \begin{cases} X'' + \lambda^2 X = 0 \\ T'' + c^2 \lambda^2 T = 0 \end{cases}$$

$$X(x) = A \cos \lambda x + B \sin \lambda x, \quad X(0) = 0 \Rightarrow A = 0$$

$$X(L) = 0 \Rightarrow B \sin \lambda L = 0 \Rightarrow \lambda L = n\pi \quad (n \in \mathbb{Z}) \Rightarrow \lambda = \frac{n\pi}{L} = \lambda_n$$

$$X_n(x) = B_n \sin \lambda_n x, \quad \lambda_n = \frac{n\pi}{L} \quad (n \in \mathbb{Z})$$

$$T'' + c^2 \lambda_n^2 T = 0 \Rightarrow T_n(t) = C_n \cos(c\lambda_n t) + D_n \sin(c\lambda_n t)$$

$$u_n(x, t) = X_n(x) \times T_n(t) = B_n \sin \lambda_n x \times (C_n \cos(c\lambda_n t) + D_n \sin(c\lambda_n t))$$

Superposing the solutions:

$$u(x, t) = \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{cn\pi}{L}t\right) + b_n \sin\left(\frac{cn\pi}{L}t\right) \right] \sin \frac{n\pi}{L}x$$

Applying the initial conditions. The condition $u(x, 0) = f(x)$:

$$u(x, 0) = \sum_{n=1}^{\infty} a_n \sin \frac{n\pi}{L}x = f(x)$$

$f(x)$ has been set equal to a periodic odd function, so to compute the unknown Fourier coefficients we must regard $f(x)$ as a periodic odd function (with period $2L$). Then

$$a_n = \frac{2}{2L} \int_{-L}^L f(x) \sin\left(\frac{2\pi}{2L}nx\right) dx = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

The condition $u_t(x, 0) = g(x)$:

$$u_t(x, t) = \sum_{n=1}^{\infty} \left[-a_n \frac{cn\pi}{L} \sin\left(\frac{cn\pi}{L}t\right) + b_n \frac{cn\pi}{L} \cos\left(\frac{cn\pi}{L}t\right) \right] \sin \frac{n\pi}{L}x$$

$$u_t(x, 0) = \sum_{n=1}^{\infty} b_n \frac{cn\pi}{L} \sin \frac{n\pi}{L}x = g(x) \Rightarrow b_n = \frac{2}{cn\pi} \int_0^L g(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Key result 11.4 ■ Solution of the one-dimensional wave equation (separation of variables)

$$u(x, t) = \sum_{n=1}^{\infty} \left[a_n \cos\left(\frac{cn\pi}{L}t\right) + b_n \sin\left(\frac{cn\pi}{L}t\right) \right] \sin \frac{n\pi}{L}x$$

$$a_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx, \quad b_n = \frac{2}{cn\pi} \int_0^L g(x) \sin\left(\frac{n\pi}{L}x\right) dx$$

Example 11.5 ■ Example 1

Find the solution of the following one-dimensional wave equation.

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = u(\pi, t) = 0, \quad u(x, 0) = k \sin 2x, \quad u_t(x, 0) = 0$$

Since the initial velocity of the vibrating string is zero, $g(x) = 0 \Rightarrow b_n = 0$. Hence ($c = 1$, $L = \pi$):

$$u(x, t) = \sum_{n=1}^{\infty} a_n \cos\left(\frac{cn\pi}{L}t\right) \sin \frac{n\pi}{L}x = \sum_{n=1}^{\infty} a_n \cos nt \sin nx$$

$$a_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi}{L}x\right) dx = \frac{2}{\pi} \int_0^{\pi} k \sin 2x \sin nx dx = \begin{cases} 0 & n \neq 2 \\ k & n = 2 \end{cases}$$

$$u(x, t) = k \sin 2x \cos 2t$$

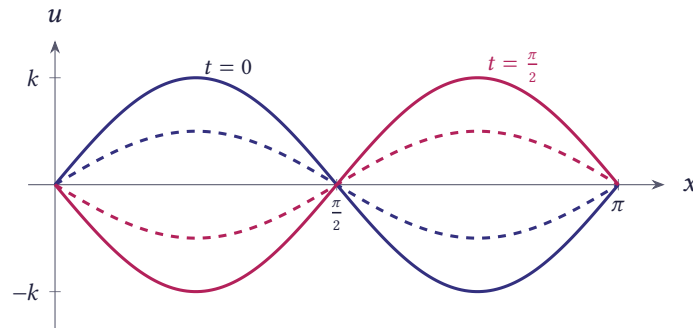


Figure 11.2 The second mode of the string: $u(x, t) = k \sin 2x \cos 2t$ at different times

Example 11.6 ■ Example 2

Find the solution of the following one-dimensional wave equation.

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = u(\pi, t) = 0, \quad u(x, 0) = f(x), \quad u_t(x, 0) = 0$$

where $f(x)$ is the triangular function: $f(x) = \frac{k}{a}x$ for $0 \leq x \leq a$ and $f(x) = \frac{k}{a-\pi}(x-\pi)$ for $a \leq x \leq \pi$.

$u_t(x, 0) = g(x) = 0 \Rightarrow b_n = 0$ and

$$u(x, t) = \sum_{n=1}^{\infty} a_n \cos nt \sin nx$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx = \frac{2}{\pi} \int_0^a \frac{k}{a} x \sin nx \, dx + \frac{2}{\pi} \int_a^{\pi} \frac{k}{a-\pi} (x-\pi) \sin nx \, dx \\ &= \frac{2k}{a\pi} \left[x \left(\frac{-1}{n} \cos nx \right) - (1) \left(\frac{-1}{n^2} \sin nx \right) \right]_0^a + \frac{2k}{(a-\pi)\pi} \left[(x-\pi) \left(\frac{-1}{n} \cos nx \right) - (1) \left(\frac{-1}{n^2} \sin nx \right) \right]_a^{\pi} \\ &= \dots = \frac{2k \sin na}{an^2(\pi-a)} \end{aligned}$$

$$u(x, t) = \frac{2k}{a(\pi-a)} \sum_{n=1}^{\infty} \frac{\sin na}{n^2} \cos nt \sin nx$$

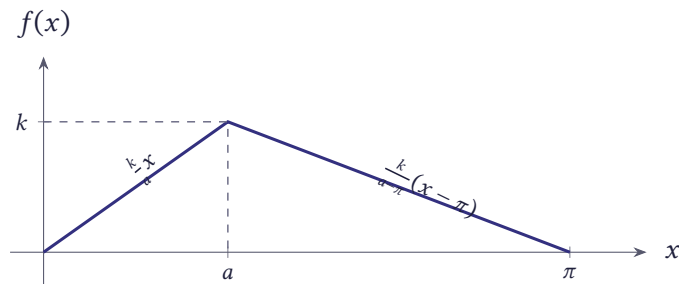


Figure 11.3 The triangular initial profile $f(x)$ with peak value k at $x = a$

Example 11.7 ■ Example 3

Find the solution of the following one-dimensional wave equation.

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = u(\pi, t) = 0, \quad u(x, 0) = kx(\pi - x), \quad u_t(x, 0) = 0$$

$g(x) = 0 \Rightarrow b_n = 0$ and

$$u(x, t) = \sum_{n=1}^{\infty} a_n \cos nt \sin nx$$

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_0^{\pi} kx(\pi - x) \sin nx \, dx \\ &= \frac{2k}{\pi} \left[(-x^2 + \pi x) \left(\frac{-1}{n} \cos nx \right) - (-2x + \pi) \left(\frac{-1}{n^2} \sin nx \right) + (-2) \left(\frac{1}{n^3} \cos nx \right) \right]_0^{\pi} = \dots \\ &= \frac{4k(1 - (-1)^n)}{\pi n^3} \end{aligned}$$

$$u(x, t) = \frac{4k}{\pi} \sum_{n=1}^{\infty} \frac{1 - (-1)^n}{n^3} \cos nt \sin nx$$

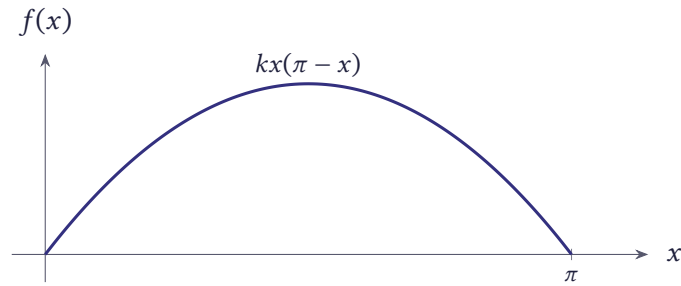


Figure 11.4 The parabolic initial profile $f(x) = kx(\pi - x)$

Exercise 11.8

- Find the ratio $\frac{a_1^2}{a_1^2 + a_2^2 + \dots}$.
- Plot the ratio a_{2n-1}/a_{2n+1} as a function of $n \in \mathbb{N}$.

11.4 The Method of Separation of Variables: Examples

Example 11.9 ■ Example 1

Solve the following differential equation by separation of variables: $xu_x = yu_y$.

$$x \frac{\partial u}{\partial x} = y \frac{\partial u}{\partial y}, \quad u(x, y) = X(x) \cdot Y(y)$$

$$xX'Y = yXY' \Rightarrow \frac{xX'}{X} = \frac{yY'}{Y} = k$$

$$\frac{X'}{X} = \frac{k}{x} \Rightarrow \ln X = k \ln x + c_1 \Rightarrow X(x) = C_1 x^k, \quad \frac{Y'}{Y} = \frac{k}{y} \Rightarrow \ln Y = k \ln y + c_2 \Rightarrow Y(y) = C_2 y^k$$

$$u(x, y) = X(x)Y(y) = C_1 x^k C_2 y^k = C(xy)^k$$

Example 11.10 ■ Example 2

Solve the following differential equation by separation of variables: $u_{xx} + u_{yy} = 0$.

$$u(x, y) = X(x) \cdot Y(y) \Rightarrow X''Y + XY'' = 0 \Rightarrow \frac{X''}{X} = -\frac{Y''}{Y} = k$$

$$\begin{cases} X'' - kX = 0 \Rightarrow X(x) = Ae^{\sqrt{k}x} + Be^{-\sqrt{k}x} \\ Y'' + kY = 0 \Rightarrow Y(y) = Ce^{\sqrt{-k}y} + De^{-\sqrt{-k}y} \end{cases}$$

$$u(x, y) = X(x)Y(y) = (Ae^{\sqrt{k}x} + Be^{-\sqrt{k}x})(Ce^{\sqrt{-k}y} + De^{-\sqrt{-k}y})$$

Example 11.11 ■ Example 3

Solve the following differential equation by separation of variables: $x^2 u_{xy} = -3y^2 u$.

$$x^2 \frac{\partial^2 u}{\partial x \partial y} + 3y^2 u = 0, \quad u(x, y) = X(x) \cdot Y(y) \Rightarrow x^2 X'Y' + 3y^2 XY = 0 \Rightarrow \frac{x^2 X'}{X} = \frac{-3y^2 Y'}{Y} = k$$

$$\frac{X'}{X} = \frac{k}{x^2} \Rightarrow \ln X = \frac{-k}{x} + c_1 \Rightarrow X(x) = C_1 e^{-k/x}$$

$$\frac{Y'}{Y} = \frac{-3y^2}{k} \Rightarrow \ln Y = -\frac{y^3}{k} + c_2 \Rightarrow Y(y) = C_2 e^{-y^3/k}$$

$$u(x, y) = X(x)Y(y) = C_1 e^{-k/x} \times C_2 e^{-y^3/k} = C e^{-(k/x + y^3/k)}$$

11.5 The One-Dimensional Wave Equation with Nonzero Boundary Values

Example 11.12 ■ Problem

Find the solution of the following one-dimensional wave equation.

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = P_0, \quad u(L, t) = P_1, \quad u(x, 0) = f(x), \quad u_t(x, 0) = g(x)$$

$$u(x, t) = w(x, t) + v(x) \implies \begin{cases} w(0, t) + v(0) = P_0 \\ w(L, t) + v(L) = P_1 \end{cases}$$

$$\frac{\partial^2 w}{\partial t^2} = c^2 \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 v}{\partial x^2} \right)$$

We choose $v(x)$ so that the problem for w has homogeneous boundary conditions:

$$\begin{cases} \frac{\partial^2 v}{\partial x^2} = 0 \\ v(0) = P_0 \\ v(L) = P_1 \end{cases} \implies v(x) = \left(\frac{P_1 - P_0}{L} \right) x + P_0$$

$$\begin{cases} \frac{\partial^2 w}{\partial t^2} = c^2 \frac{\partial^2 w}{\partial x^2} \\ w(0, t) = w(L, t) = 0 \end{cases} \implies w(x, t) = \sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{cn\pi}{L}t\right) + B_n \sin\left(\frac{cn\pi}{L}t\right) \right] \sin \frac{n\pi}{L}x$$

$$u(x, t) = \underbrace{\sum_{n=1}^{\infty} \left[A_n \cos\left(\frac{cn\pi}{L}t\right) + B_n \sin\left(\frac{cn\pi}{L}t\right) \right] \sin \frac{n\pi}{L}x}_{w(x, t)} + \underbrace{\left(\frac{P_1 - P_0}{L} \right) x + P_0}_{v(x)}$$

The initial condition $u(x, 0) = f(x)$:

$$\sum_{n=1}^{\infty} A_n \sin \frac{n\pi}{L}x = f(x) - \left(\frac{P_1 - P_0}{L} \right) x - P_0 \triangleq f_1(x)$$

$$A_n = \frac{2}{L} \int_0^L f_1(x) \sin \frac{n\pi}{L}x \, dx = \frac{2}{L} \int_0^L \left[f(x) - \left(\frac{P_1 - P_0}{L} \right) x - P_0 \right] \sin \frac{n\pi}{L}x \, dx$$

Exercise 11.13

Find B_n in the cases (a) $g(x) = 0$ and (b) $g(x)$ arbitrary.

11.6 The One-Dimensional Wave Equation with Zero Initial Velocity (Travelling Waves)

If the initial velocity is zero ($g(x) = 0$), then

$$u(x, t) = \sum_{n=1}^{\infty} a_n \cos\left(\frac{cn\pi}{L}t\right) \sin \frac{n\pi}{L}x, \quad u(x, 0) = \sum_{n=1}^{\infty} a_n \sin \frac{n\pi}{L}x = f(x) \equiv \tilde{f}(x)$$

Using $\sin \alpha \cos \beta = \frac{1}{2}(\sin(\alpha + \beta) + \sin(\alpha - \beta))$:

$$\begin{aligned} u(x, t) &= \sum_{n=1}^{\infty} a_n \frac{1}{2} \left[\sin\left(\frac{n\pi}{L}x + \frac{cn\pi}{L}t\right) + \sin\left(\frac{n\pi}{L}x - \frac{cn\pi}{L}t\right) \right] \\ &= \frac{1}{2} \sum_{n=1}^{\infty} a_n \left[\sin \frac{n\pi}{L}(x + ct) + \sin \frac{n\pi}{L}(x - ct) \right] \\ &= \frac{1}{2} \sum_{n=1}^{\infty} a_n \sin \frac{n\pi}{L}(x + ct) + \frac{1}{2} \sum_{n=1}^{\infty} a_n \sin \frac{n\pi}{L}(x - ct) = \frac{1}{2} [\tilde{f}(x + ct) + \tilde{f}(x - ct)] \end{aligned}$$

Key result 11.14 ■ Travelling waves

$$u(x, t) = \frac{1}{2} [\tilde{f}(x + ct) + \tilde{f}(x - ct)]$$

$\tilde{f}$ is the odd periodic extension (with period $2L$) of the initial function f ; the solution is the superposition of two waves of half the initial amplitude travelling at speed c in opposite directions.

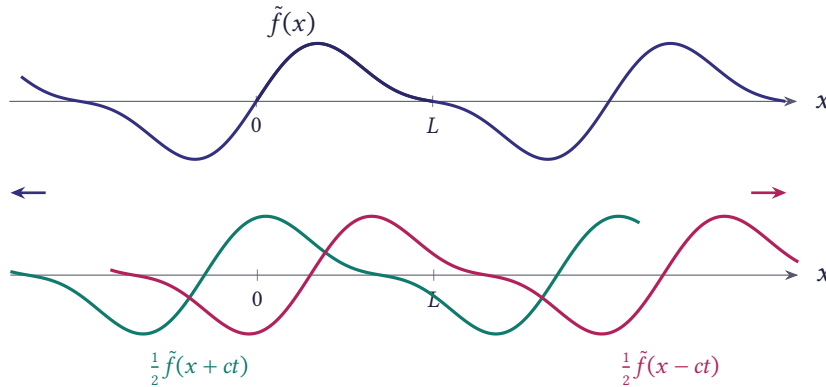


Figure 11.5 The odd periodic extension $\tilde{f}(x)$ and the two waves $\frac{1}{2}\tilde{f}(x + ct)$ and $\frac{1}{2}\tilde{f}(x - ct)$ travelling in opposite directions

Example 11.15

Plot the solution of the following one-dimensional wave equation, assuming zero initial velocity, at different times.

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \quad u(0, t) = u(L, t) = 0, \quad u(x, 0) = f(x), \quad u_t(x, 0) = 0$$

$$f(x) = \begin{cases} \frac{2k}{L}x & 0 < x < \frac{L}{2} \\ \frac{2k}{L}(L-x) & \frac{L}{2} < x < L \end{cases}$$

Plotting $\frac{1}{2}\tilde{f}(x+ct)$ and $\frac{1}{2}\tilde{f}(x-ct)$ and adding them at the times $t = 0, t = L/5c, t = 2L/5c, t = L/2c$ and $t = 3L/5c$ gives the shape of the solution.

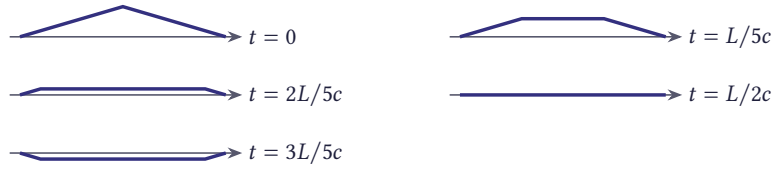


Figure 11.6 The solution of the example at $t = 0, L/5c, 2L/5c, L/2c$ and $3L/5c$; at $t = L/2c$ the string is flat